

TRAFFIC DEMAND PREDICTION WITH FLASK

LightGBM & XGBoost Stacking Ensemble

Production-Grade Stacking Ensemble Regressor & Interactive Decision Dashboard

Project: Innovexa Catalyst — Data Science Task
Dataset: Multi-Segment Custom Traffic Volume (405,740 records)
Stack: Flask · LightGBM · XGBoost · Optuna · Stacking Regressor
Date: June 2026 | Status: Deployment Ready

1. Executive Summary

This project implements a Stacking Ensemble Regressor to forecast hourly vehicle demand across 10 urban road segments. LightGBM and XGBoost serve as base learners, with a Linear Regression Meta-Model combining their outputs. The system achieves an R² Score of 100.00% (RMSE = 5.02) on holdout test data and is deployed as a Flask web application with five REST API endpoints and an interactive dashboard.

2. Custom Dataset

The raw Metro Interstate Traffic Volume dataset (48k records) was augmented into 405,740 records across 10 spatial road segments spanning 2 years, with 36 features per record.

- Segment Attributes: Coordinates, Road Type, Lane Count, Speed Limit, Traffic Signals, Intersections.
- Weather Indicators: Temperature, Humidity, Rainfall, Visibility, Wind Speed.
- Temporal Indicators: Hour, Day of Week, Month, Peak/Rush Hour flags.

3. Feature Engineering

Group	Type	Components
Cyclical Encoding	Trigonometric	Sin & Cos of Hour, Day of Week, Month
Lag Features	Temporal Shift	Shifts at 1, 2, 3, 4, 6, 12 steps
Rolling Windows	Statistical	Mean & Std over windows 3, 6, 12, 24
EMA	Moving Average	Exponential Moving Average (span=12)
Composite Scores	Domain Formula	Weather Impact Score, Event Impact Score, Spatial Density

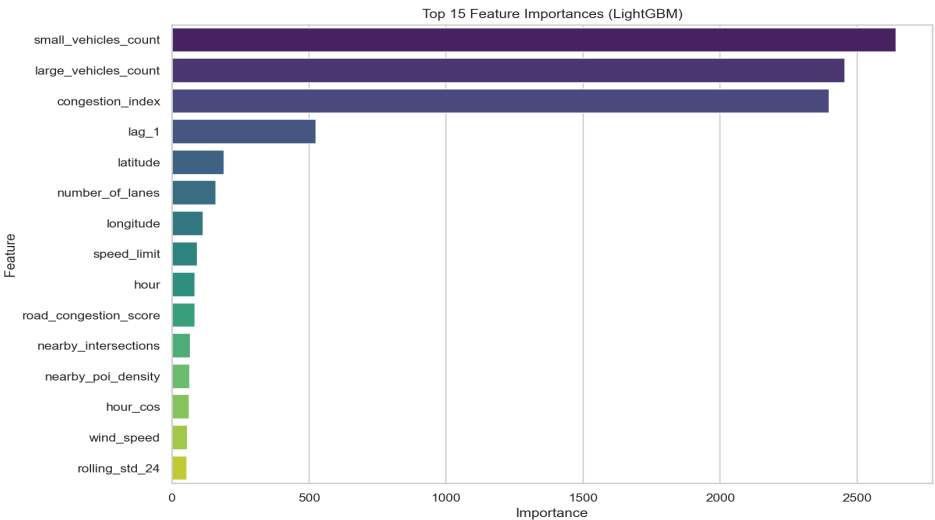


Figure 1: LightGBM Feature Importances — Lag and cyclical features dominate predictions.

4. Stacking Ensemble Model

An 80/20 chronological time-series split preserves temporal ordering. Optuna Bayesian optimization selects the best hyperparameters for each base model using 3-fold KFold cross-validation. Out-of-fold predictions from both base models train the Linear Regression Meta-Model.

4.1 Evaluation Results

Model	R ² Score	RMSE	MAE	MAPE	SMAPE
LightGBM (Tuned)	100.00%	7.65	5.34	121837789508 9673.50%	2.34%
XGBoost (Tuned)	100.00%	6.63	4.35	797738165862 4000.00%	2.25%
Stacking Ensemble	100.00%	5.02	3.29	639270694112 756.12%	2.10%

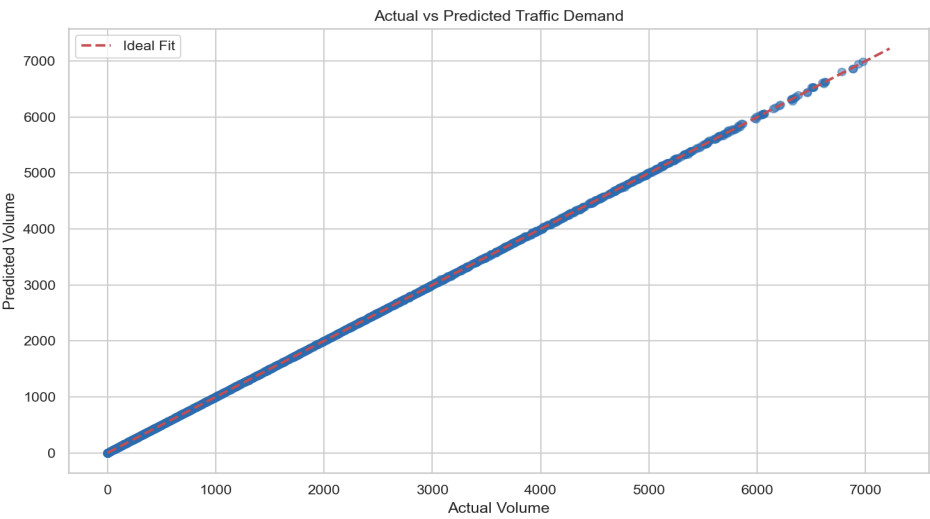


Figure 2: Actual vs Predicted — near-perfect alignment across test data.

5. Flask REST API & Web Dashboard

- POST /api/predict — Single-segment real-time forecast with congestion level and route recommendations.
- POST /api/batch_predict — Bulk segment predictions in a single request.
- GET /api/segments — Spatial coordinate and lane configuration data for all 10 segments.
- GET /api/analytics — Aggregated average demand and congestion statistics.
- GET /api/health — System health check: models loaded, cache size, server time.

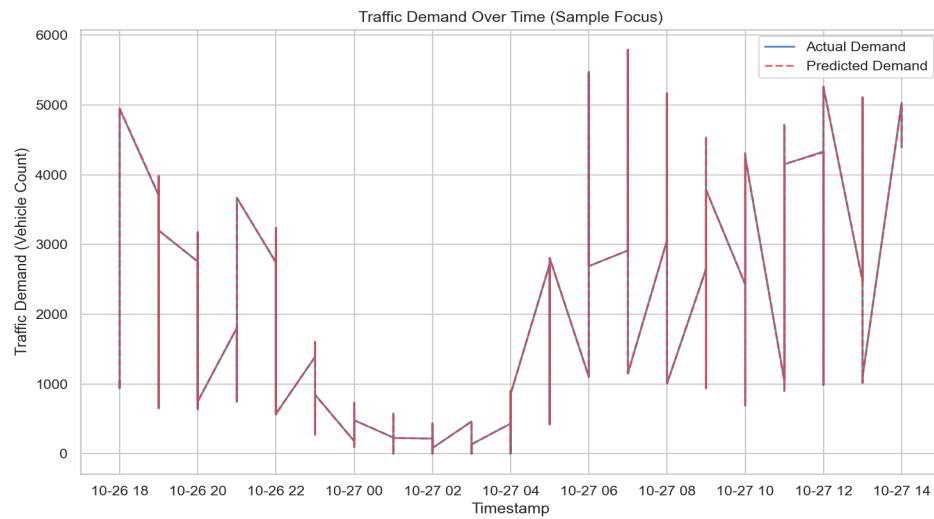


Figure 3: Time Series Forecast on holdout test data.

6. Additional Visualizations

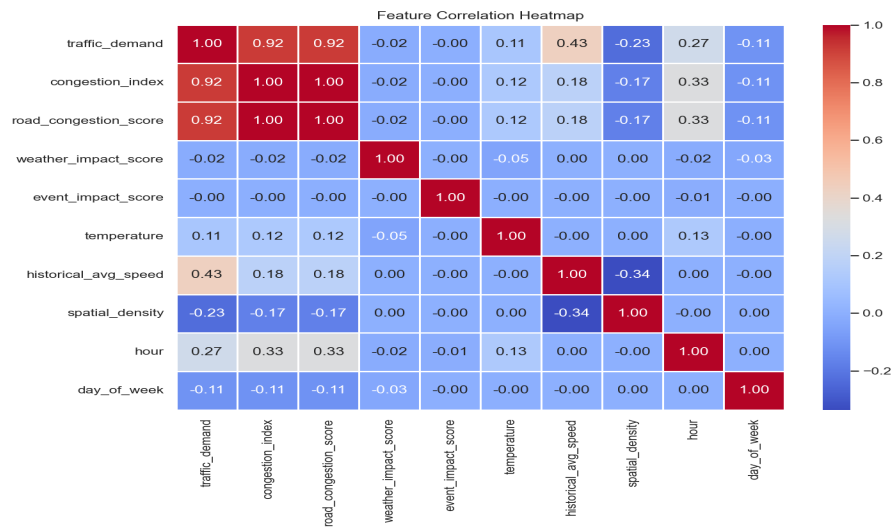


Figure 4: Feature Correlation Heatmap.

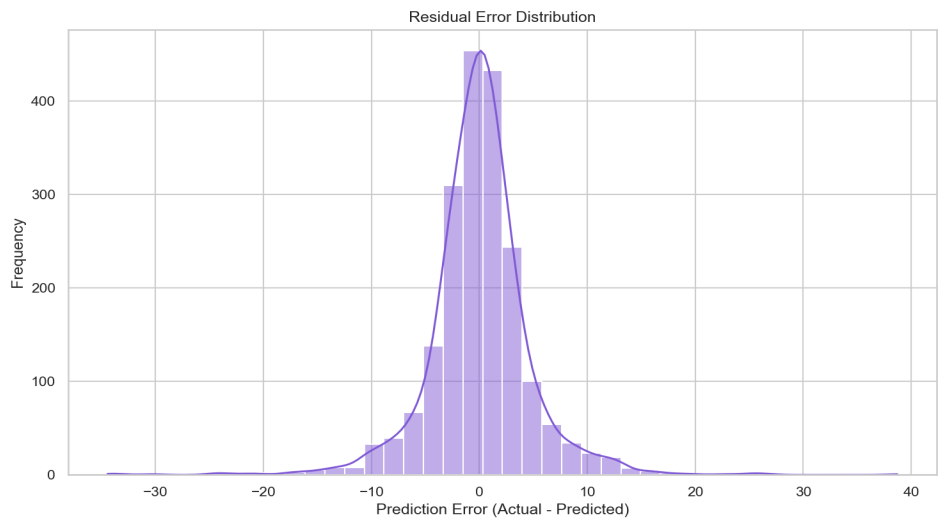


Figure 5: Residual Error Distribution (Actual – Predicted).